Quick-start on Argus-RCE Flow

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一站式EDA及相关服务提供商

Argus-RCE Flow

- Argus-RCE GUI-Flow
  - Run RCE Parasitic Extraction for GUI
- Argus-RCE CUI-Flow
  - Run RCE command line file for CUI

Before Running Argus-RCE

- It is recommended to execute parasitic extraction based on a LVS-clean database. Therefore, the required input files include the followings,
  - The CDL / SPICE netlist of your design
  - The corresponding layout GDS
  - Argus LVS runset

- Generally, there are 2 parts to realize Argus-RCE GUI flow:
  - Part I, setup Argus LVS runset
  - Part II, start Argus-RCE GUI

Part I: Argus Runset Settings

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- Setup **metallization options** you preferred, and open **RC switch**, such as:

#DEFINE METAL_OPTION	(Define the metal layers)
#DEFINE RC_RUN	(Define RC RUN **TRUE**)
#DEFINE ENABLE_PDSP	(Define ENABLE PDSP **TRUE**)
#DEFINE Back_Annotation_Flow	(Define the extraction flow you preferred)

- The following are the example of Argus runset settings,

```
#DEFINE METAL_OPTION 10

//******************************************************************************
//*OPTION 2: Define RC RUN or not.The value can be TRUE or FALSE(Upper Case) .
//*TRUE : It is used for LVS+RC post-simulation flow.
//*FALSE: It is used for LVS check.
//*Default value is FALSE.

#DEFINE RC_RUN TRUE
//******************************************************************************
//*OPTION 3: Define the back annotation flow.The value can be 1 or 2.
//*1: Extrview flow,LVS will reserve some extra parameters for back annotation.
//*2: Netlist flow,LVS will not reserve the back annotation parameters.
//*Default value is 2.

#DEFINE Back_Annotation_Flow 2
//******************************************************************************
//*OPTION 4: Define the soft connectivity report in file "lvs.report".The value can be TRUE or FALSE
//*TRUE :  It may be lead to lvs_report db error, LVS will fail.
//*FALSE: It does not effect lvs result,but will report warning in the file *.softchk,
//         so you must check this file *.softchk and fix the soft connectivity warning.
//* Default value is FALSE.

#DEFINE ENABLE_PDSP TRUE
//******************************************************************************
```

Part II: Run Argus-RCE GUI

- **STEP 1**: Confirm pdk library path defined in **lib.defs**. If you want to run back-annotation, you must also define the anaologLib or amsLib, such as:

  DEFINE pdkLib ./pdkLib

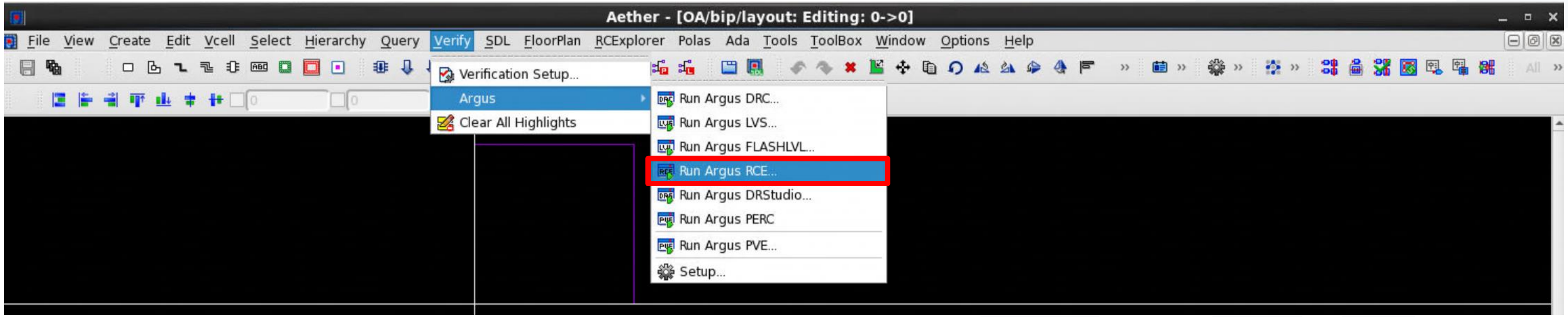
  ASSIGN pdkLib libMode shared

  DEFINE amsLib ./amsLib

  ASSIGN amsLib libMode shared

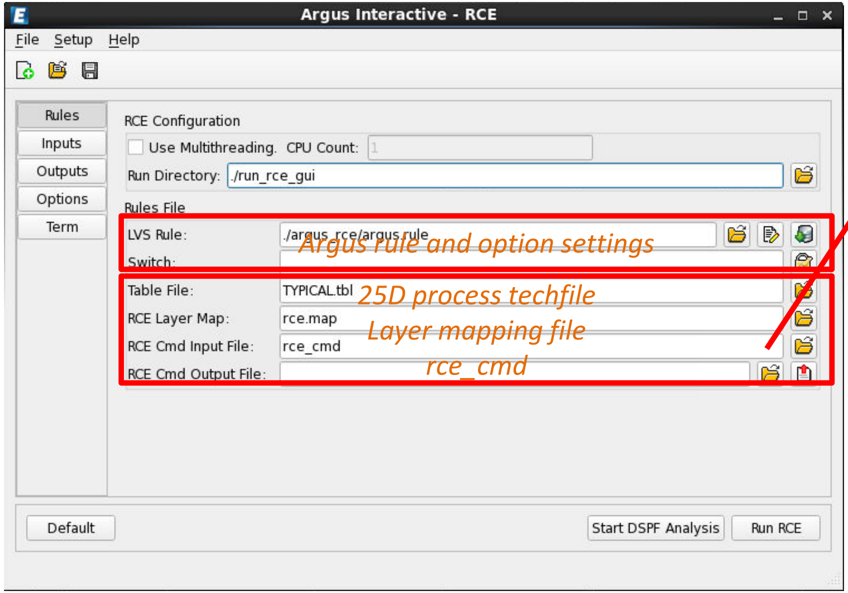
- **STEP 2**: Start Aether, click

  Verify->Argus->Run Argus RCE…



Part II: Run Argus-RCE GUI (Cont.)

- **STEP 3**: Setup Argus Interactive – RCE,
  - Rules menu

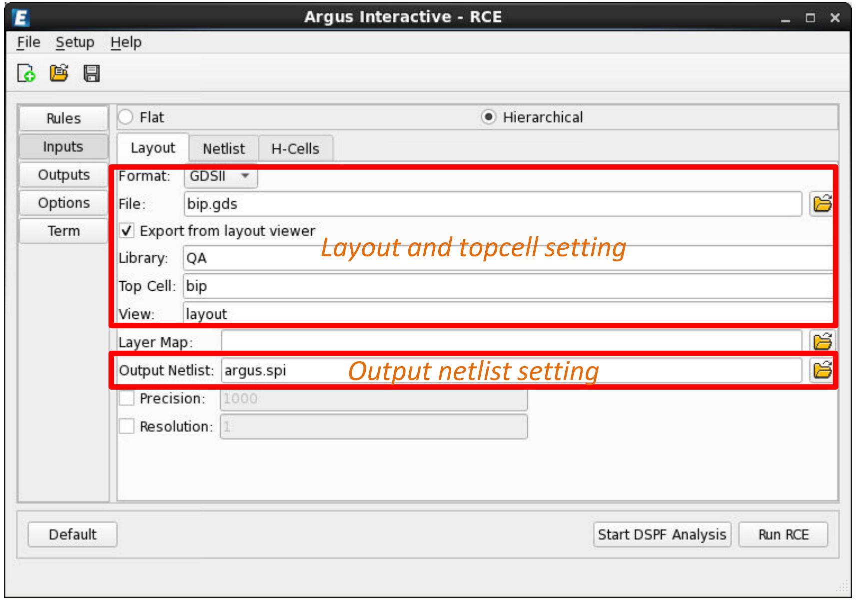


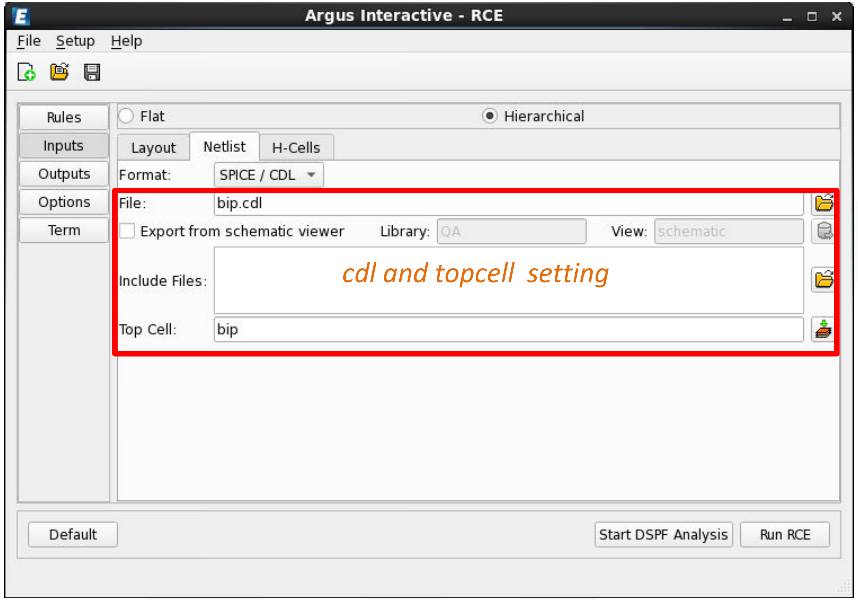
NOTE:
If you want to run back-annotation, please set OUTPUT_EXT_VIEW : true, and specify EXTVIEW_CONFIG in **rce_cmd**, such as,

```
###########Extract View Setup###########
OUTPUT_EXT_VIEW : true
EXT_VIEW_LIB :
EXT_VIEW_SLIB :
EXT_VIEW_SCELL :
EXT_VIEW_CDL :
EXTVIEW_CELLMAP :
EXTVIEW_CONFIG : extract_view.file
EXTVIEW_BLOCK_CELL_VIEW :
EXTVIEW_R_MODEL_LIB : amsLib
EXTVIEW_R_MODEL_CELL : presistor
EXTVIEW_R_MODEL_VIEW : symbol
EXTVIEW_C_MODEL_LIB : amsLib
EXTVIEW_C_MODEL_CELL : pcapacitor
EXTVIEW_C_MODEL_VIEW : symbol
```

Part II: Run Argus-RCE GUI (Cont.)

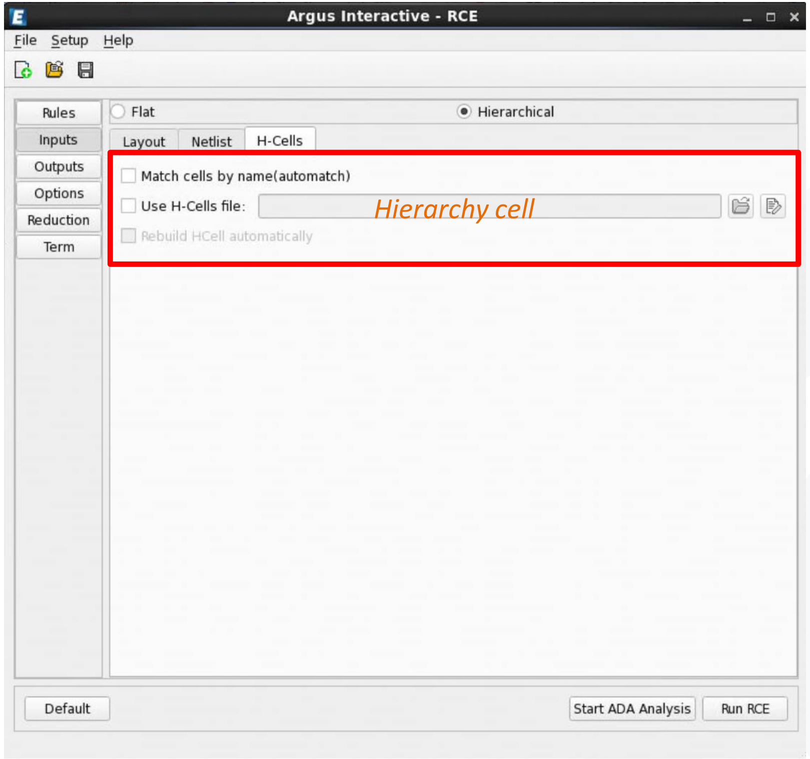
- **STEP 3**: Setup Argus Interactive – RCE,
  - Input menu (Layout)
  - Input menu (Netlist)





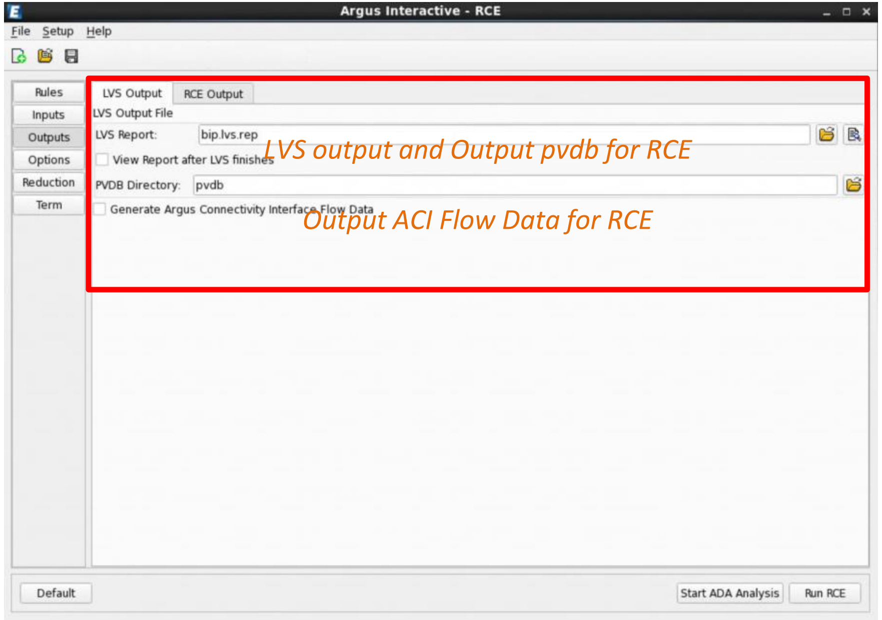
Part II: Run Argus-RCE GUI (Cont.)

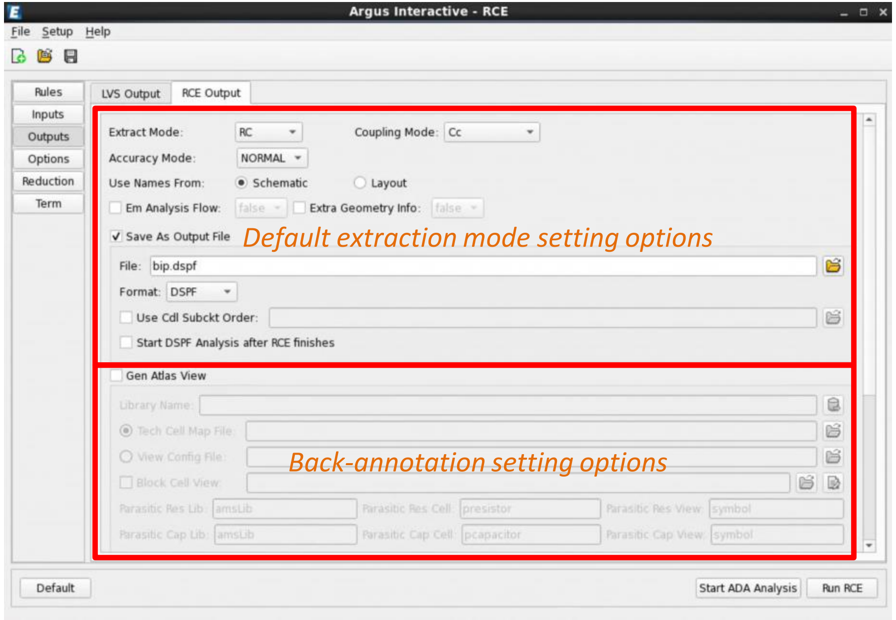
- **STEP 3**: Setup Argus Interactive – RCE,
  - Input menu (H-Cells)



Part II: Run Argus-RCE GUI (Cont.)

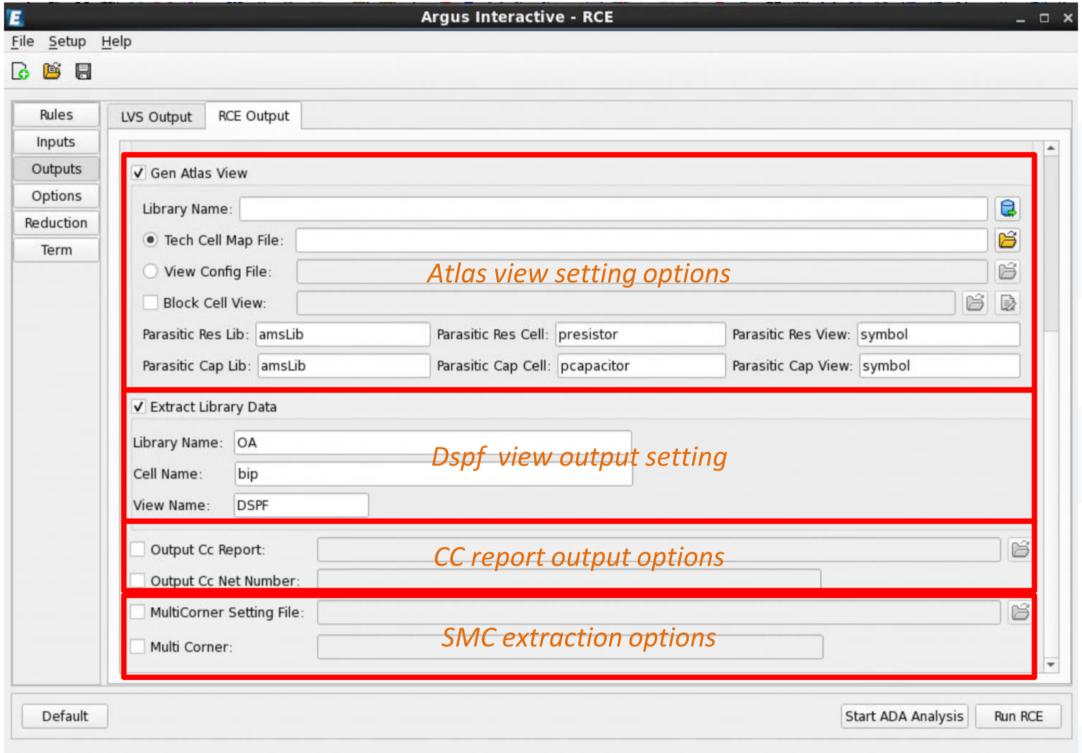
- **STEP 3**: Setup Argus Interactive – RCE,
  - Outputs menu





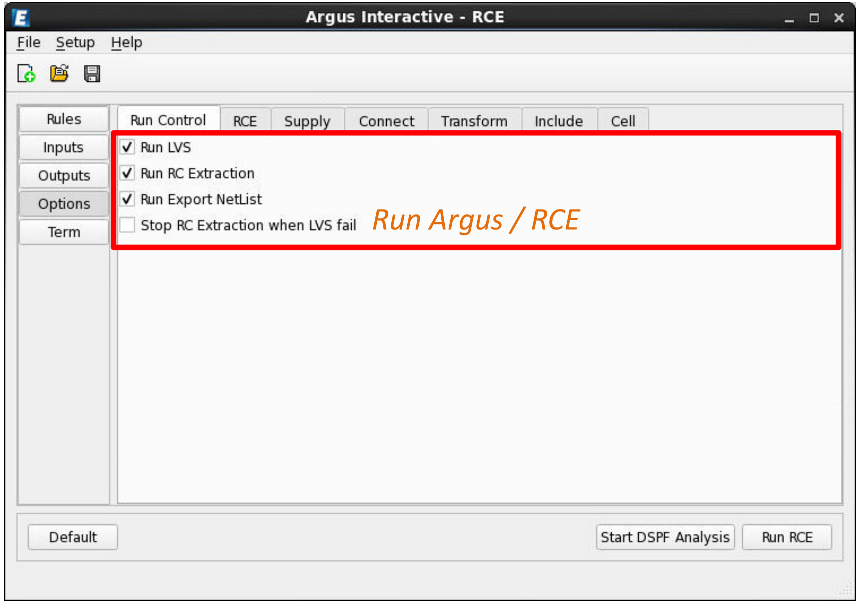
Part II: Run Argus-RCE GUI (Cont.)

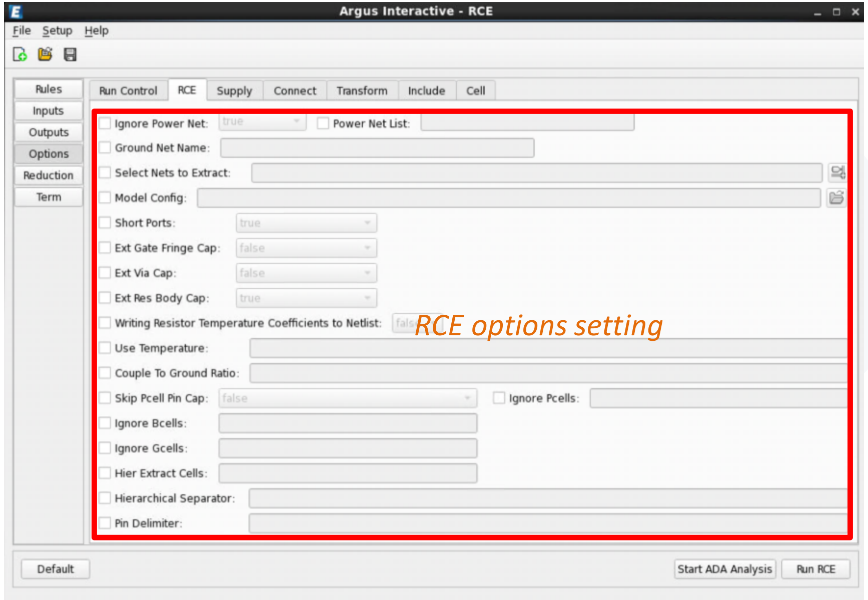
- **STEP 3**: Setup Argus Interactive – RCE,
  - back-annotation menu



Part II: Run Argus-RCE GUI (Cont.)

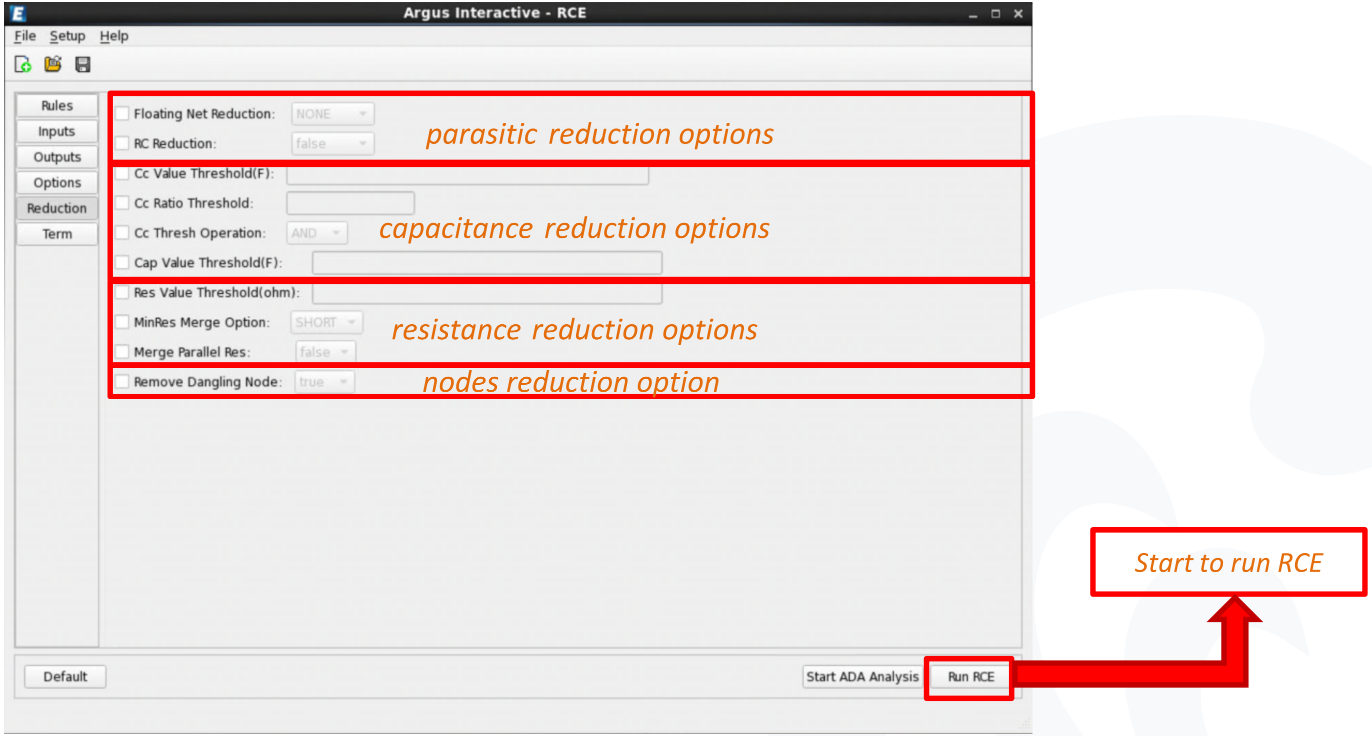
- **STEP 3**: Setup Argus Interactive – RCE,
  - Options menu





Part II: Run Argus-RCE GUI (Cont.)

- **STEP 3**: Setup Argus Interactive – RCE,
  - Reduction menu



Argus-RCE Flow

➢ Argus-RCE GUI-Flow

- Run RCE Parasitic Extraction for GUI

➢ Argus-RCE CUI-Flow

- Run RCE command line file for CUI

Before Running Argus-RCE

- It is recommended to execute parasitic extraction based on a LVS-clean database. Therefore, the required input files include the followings,
  - The CDL / SPICE netlist of your design
  - The corresponding layout GDS
  - Argus LVS runset

- Generally, there are 3 parts to realize Argus-RCE GUI flow:
  - Part I, setup Argus LVS runset, run Argus LVS to output pvdb as RCE input:

    argus –lvs *[-flat]* ***–rcpvdb*** *argus.lvs* *or*

    argus –lvs –hier –rcpvdb –hcell *hcell_list argus.rule*
  - Part II, translate Argus pvdb to OA format library for RCE:

    pvdb2oa –pvdb *pvdb* ***–l*** *topcell_oaLib* ***–nl –termshape –lic RCE_DBC –lvs***
  - Part III, setup rce_cmd, run RCE to do parasitic extraction:

    empExtract *rce_cmd*

Part I: Argus Runset Settings

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- Firstly, setup **metallization options** you preferred, open **RC switch**, such as:
  - *#DEFINE METAL_OPTION* (Define the metal layers)
  - *#DEFINE RC_RUN* (Define RC RUN **TRUE**)
  - *#DEFINE ENABLE_PDSP* (Define ENABLE PDSP **TRUE**)
  - *#DEFINE Back_Annotation_Flow* (Define the extraction flow you preferred)

- Secondly, specify **LAYOUT** , **SCHEMATIC** respectively, such as:
  - *layout_input (path "***.gds")* (Specify layout path)
  - *layout_input (top_cell "***")* (Specify layout topcell name)
  - *schematic_input (path "***.cdl")* (Specify schematic path)
  - *schematic_input (top_cell "***")* (Specify schematic topcell name)

Part I: Argus Runset Settings (Cont.)

- The following are the example of Argus runset settings

```
#DEFINE METAL_OPTION 10

//****************************************************************************
//*OPTION 2: Define RC RUN or not.The value can be TRUE or FALSE(Upper Case) .
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//****************************************************************************
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#DEFINE Back_Annotation_Flow 2
//****************************************************************************
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//*TRUE :  It may be lead to lvs_report db error, LVS will fail.
//*FALSE: It does not effect lvs result,but will report warning in the file *.softchk,
//        so you must check this file *.softchk and fix the soft connectivity warning.
//* Default value is FALSE.

#DEFINE ENABLE_PDSP TRUE
//****************************************************************************
layout_input ( path "./bjt.gds"  )
layout_input ( top_cell "bjt"  )

schematic_input ( path "./bjt.cdl"  )
schematic_input ( top_cell "bjt"  )

layout_input ( format gds  )
schematic_input ( format spice  )
```

- After Argus runset properly, following cmd need to be executed for LVS

  ***argus –lvs** [-flat] **–rcpvdb** argus.lvs* *or*

  ***argus –lvs –hier –rcpvdb –hcell** hcell_list argus.rule*

Part II: pvdb2oa

- After LVS, Argus would create a directory named as pvdb , we need to transfer it to OA format library by:

  pvdb2oa –pvdb *pvdb* ***–l*** *topcell_oaLib* ***–nl –termshape –lic RCE_DBC –lvs***

  Where *topcell_oaLib* is the output OA library need by next RCE extraction

Part III: RCE cmd setting

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- After we get *topcell_oaLib*, we need prepare an RCE extraction option file named **rce_cmd**, this file usually provided in released TAR package. Users can modify these settings based on own design. There are some critical options setting provided as example,

```
###########Layout Data Setup###########
TOP_CELL : topcell

#----------Oa Option----------
OALIB : topcell_oaLib
VIEW : layout
#----------CCX Option----------
QUERY_FILE :
TECH_FILE : ***.etf
LAYER_MAP : ***.map
TABLE : ***.tbl
##########Output Setup############
NETLIST_FORMAT : dspf  #(dspf/spef)
NETLIST_ORIGINAL_COOR : true
NETLIST_INSTANCE_SECTION : true  #(all/true/false)
NETLIST_NODENAME_NETNAME : true
NETLIST_FILE : ***.dspf
NETLIST_GROUND_NODE_NAME : 0
COUPLE_TO_GROUND_RATIO : 1
PARALLEL_R_MERGE : false
CAP_VALUE_THRESH : -1
RES_VALUE_THRESH : -1
MINRES_MERGE_OPTION : SHORT  #(SHORT/MERGE)
#OUTPUT_3DVIEW_FILE :
OUTPUT_CC_REPORT : ***.cc.rpt
OUTPUT_CC_NET_CNT :
RCMODE : RC  #(RC/R/C/FS_COMPARE/NORC)
```

Specify input oa and topcell name

Specify layermap and 25D process table

Specify output netlist

Specify output cc report

RC Extraction mode

```
###########Extract View Setup###########
OUTPUT_EXT_VIEW : false
EXT_VIEW_LIB :
EXT_VIEW_SLIB :
EXT_VIEW_SCELL :
EXT_VIEW_CDL :
EXTVIEW_CELLMAP :
EXTVIEW_CONFIG : ***extract_view.file
EXTVIEW_BLOCK_CELL_VIEW :
EXTVIEW_R_MODEL_LIB : amsLib
EXTVIEW_R_MODEL_CELL : presistor
EXTVIEW_R_MODEL_VIEW : symbol
EXTVIEW_C_MODEL_LIB : amsLib
EXTVIEW_C_MODEL_CELL : pcapacitor
EXTVIEW_C_MODEL_VIEW : symbol
###########PCell Setup###########
IGNORE_PCELLS : pvar08 ckt* pvar18 ckt* mim ckt*  mom prim do not use*
IGNORE_BCELLS :
IGNORE_GCELLS :
SKIP_PCELL_PIN_CAP : false  #(true/false)

###########Target Setup###########
NETLIST_SELECT_NETS :
TARGET_NETS : *
TARGET_NET_FILE :
FS_CAP_NETS :
FS_CAP_NET_FILE :
POWER_NETS :
SKIP_POWER_NETS : true  #(true/fales/RES_ONLY/DEV_LAYERS)
########### Flow    Param Setup###########
THREAD_CNT : 4
CASE_SENSITIVE : true
```

*If you want to run **back-annotation**, pls set OUTPUT_EXT_VIEW : true, and specify EXTVIEW_CONFIG*

Blocking devices

2.5D RC extraction flow

3D RC extraction flow

Multi thread mode

- Run RCExplorer to do parasitic extraction:

  ***empExtract** rce_cmd*

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